Remote Sensing Image Super-Resolution Using Deep Convolutional Neural Networks and Autoencoder

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Abstract. Deep convolutional neural networks have become the most used architectures in learning-based super-resolution methods. They can extract the mapping relationship between low-resolution images and high-resolution ones. This paper explores deep convolutional auto-encoder-based super-resolution, which includes one auto-encoder unit with several convolutional blocks in both the encoding and the decoding sub-units. In this study, we used RGB slices composed of the red, green, and blue bands of Sentinel 2B MSI images with a spatial resolution of 10 m. After degrading their resolution to restore their initial resolution, we introduced them into the model. The model's performance is encouraging (PSNR: 31.94, SSIM: 0.906, MSE: 0.00637training accuracy: 0.9055, validation accuracy: 0.90, and test accuracy: 0.89). Compared to other advanced methods such as SRCNN and VDSR for the same data, our proposed method achieves satisfactory results.

Keywords: Convolutional neural network · DCASR · Deep learning · Auto-encoders · Image super-resolution · Satellite images

1 Introduction

Remote sensing technology enables humanity to acquire information on the Earth's surface for significant purposes such as weather, forestry, agriculture, surface changes, and biodiversity. The publicly available spatial remote sensing products have a meter or several meters spatial resolution. Such a resolution does not fulfil the increasing demand for high-resolution images and precise applications. In addition, purchasing commercial products of the sub-meter resolution is costlier, especially for larger areas. Image super-resolution (SR) consists of improving an image's resolution by restoring high-resolution images from one or more corresponding low-resolution images. In other words, increasing the number of pixels in the image while retaining finer spatial details than those in the original image. It is an important tool in computer vision and image processing. It is widely used in real-world applications such as medical imaging for diagnosis, monitoring, and satellite imaging, among others. In particular, it can also provide better visual experience for watching movies and playing video games (A. Lugmayr, M. Danelljan, and et al. 2021) [24].

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Moreover, the great development of deep learning (DL) techniques has allowed the creation of models to deal with the problems of super-resolution of images and videos (G, D. Zhu et al. 2021) [26].

As a result, ANN architecture has improved from simple linear perceptron classifiers to deeper architectures called deep neural networks (DNNs), which can handle more complex models that can extract more information from the image than ANN. Today (DNNs) have demonstrated their high performance in solving super-resolution tasks (W. Peijuan et al. 2022) [25].

Image super-resolution (SR) provides a class of technics to overcome the spatial resolution limitation of sensors (J. Yang et al. 2010) [1]. These technics can enhance the image's quality resolution by transcending the diffraction limit of the sensor or the lens. In this case, optical SR proceeds by multiplexing the spatial-frequency bands. Geometric SR is used when an image is degraded with noise or blurring artifact (Chen, et al. 2022) [27].

Literature regroups SR methods into three classes (Lu et al. 2015) [14]: interpolation-based, reconstruction-based, and learning-based approaches. In this paper, we are interested in the last category. Learning-based image SR methods (Zhengfeng Shao et al. 2019) [3] proceed by finding the relationship between a low- resolution and its corresponding high-resolution image. They include deep learning- based (DLSR) or neighborhood-embedding methods for such a task. In DLSR, SR uses informative features detected by the deep network structure (e.g., convolutional neural network). The only complication of those methods resides in finding the best network architecture and supplying sufficient training samples.

This work contributes by proposing a customized architecture of deep convolutional auto-encoder-based SR (DCASR) for metric resolution satellite images. Deep learning (DL) is a breakthrough in machine learning technology (Yang et al. 2019) [4]. DL applications demonstrated a robust performance that outperformed other machine learning tools. Nowadays, one can refer to the literature to see how DL-based SR has been actively explored (e.g., Dong et al. 2014; Kim et al. 2016) [5, 6]. The architecture of the proposed model is relatively simple compared to what other studies have proposed (zhu et al. 2021) [13], containing a series of convolution blocks alternating with max-pooling operation in both the encoding and the decoding part.

In this case study, we use an RGB image composed of the red, green, and blue bands of the Sentinel 2B MSI image. The image slices undergo resolution degradation to create low-resolution (LR) images. Since auto-encoder methods are unsupervised, we feed the proposed DCASR model with a subset of LR images as training samples. Another two subsets are reserved for validation and test steps. We evaluated our model in terms of PSNR, SSIM, and accuracy metrics, then compared our results against two state-of-the-art super-resolution methods: SRCNN [20] and VDSR [19].

The model showed a high accuracy: 0.90, 0.90, and 0.89 for, respectively, training accuracy, validation accuracy, and test accuracy. Furthermore, a higher PSNR value indicates better image reconstruction quality.

2 Material and Methods

2.1 Data Preparation

This work uses data from the Sentinel 2 satellite of the European Space Agency. The Multi-Spectral Instrument (MSI) sensor provides 13 bands of different resolutions from 10 to 60 m, covering the visible, shortwave infrared (SWIR), and near-infrared (NIR) (Table 1). The images are available on the USGS Earth Explorer platform.

For This Study, we downloaded the tile number T30STC captured above Meknes city, Morocco. The cloud cover of the scene is at 2.40330%. (Table 1) presents more details about the scene for reproduction. We have used only 3 of the13 bands, Red, Green, and Blue (RGB), with a resolution of 10 m (Table 2), on which we have applied atmospheric and radiometric corrections. After that, we clipped a scene of (10793, 10793), then we split our RGB image into small images of size (251, 251) (Fig. 1). Our data set counts for 1849 sample images.

In order to increase the volume of data, we used a mirror method, a data augmentation tool that generates a mirror-like twin of an original image. With this procedure, we have 3698 images. The model is trained using 2898 samples, with 400 samples reserved for testing and another 400 for validation.

Table 1. Sentinel 2A MSI image characteristics.

Band name	Central wavelength(nm)
Entity ID	L1C_T30STC_A020426_20210202T110316
Tile Number	T30STC
Platform	SENTINEL-2B
Pixel Depth	32 Bits
Cloud Cover	2.40330

Table 2. Corresponding band characteristics of Sentinel 2A MSI.

Band name	Resolution	Central wavelength(nm)
Blue	10	492.1
Green	10	559.0
Red	10	665.0

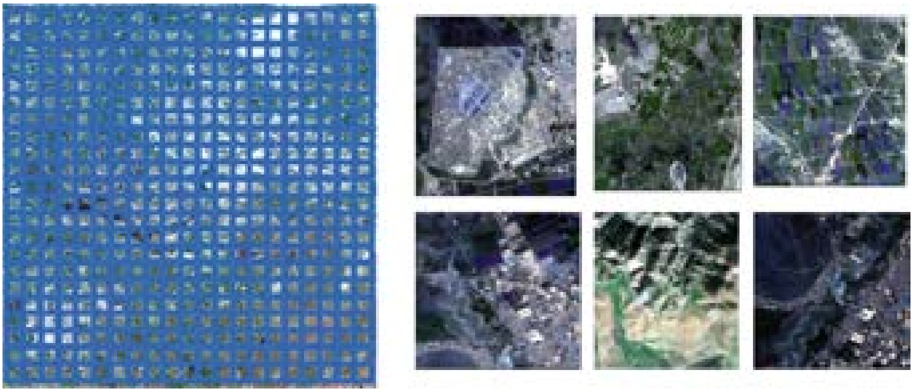

Fig. 1. In the left the filter used for clipping and in the right the clipped image examples.

2.2 Convolutional Neural Network

Convolutional neural networks (CNN) are a type of deep neural network that have proved their performance in a variety of tasks in computer vision fields such as Object detection, Face recognition, pan-sharpening [15–17], image classification [18], and image super-resolution (SR) [5]. CNN is a mathematical assembly of four types of convolutional layers, pooling, activation function and fully connected layer. The first two, convolution and pooling layers, perform feature extraction, while the last, a fully connected layer, maps extracted features into the final output.

CNN consists of several convolution layers. Each layer is made up of a set of filters (called kernels) that are applied to an input image. For each layer we define an activation function to introduce nonlinearity into the network by which they learn and approximate continuous and complex relationships between variables. Several activation functions exist, e.g. Relu, Softmax, tanH, and Sigmoid. The output of the convolutional layer is a feature map, which contains the most important features of a given input image in a process called Feature extraction.

In this case, we built five different CN layers, distributed in the DCASR architecture and sometimes repeated, with a doubling number of filters (64, 128, 256, 512, and 1024) with a stride of 1. All the filters are of the same size (3, 3). We defined the same activation function the Rectified Non-Linear Unit (ReLU) for all the CN layers due to its simplicity and performance in combating the vanishing gradient problem. The ReLU function formula is as follows:

$$\mathrm{ReLU}(x) = \max(0, x).$$

Pooling Layers are typically used after a Convolutional Layer. They aim to reduce the size of the convolved feature map before it is fed into a fully connected layer. This will reduce the training cost. There are two main types of pooling operations: max pooling and average pooling. In our case, the max-pooling layer is used to extract the maximum value from each feature map.

The fully connected layer is the final layer of a convolutional neural network (CNN). Every neuron in a fully-connected layer is fully connected to every neuron in the previous layer. As a result, it can be computed using a matrix multiplication followed by a bias affecting the fully connected layer assists in mapping the representation between the input and output.

2.3 Autoencoders

Autoencoders [22] are a type of feed forward generative model used for unsupervised learning. They aim to learn how to compress and encode data and then reconstruct the output from this reduced encoded representation to get an output that is identical to the input and has the same dimensionality. DCASR architecture (Fig. 2) consists of three parts:

The encoder: is responsible for compressing the input into a lower-dimensional representation. In this work, the input image that has 256 × 256 pixels is processed by five CNN blocks. The first block contains two convolution layers; each one defines 64 filters sized 3 × 3 and ReLU activation function. The convolution layer is followed by a max-pooling layer. The block ends with a dropout layer to reduce the overfitting. The other four blocks are configured similarly but lack the dropout layer. Each block doubles the previous block's number of filters. The output is a feature map, which is a compressed image size of 8 × 8.

The hidden layer: also known as the bottleneck, is a critical component of the network smaller than the input layer and the decoder, which contains the compressed input. We defined a convolutional layer with the same configuration and 1024 filters.

The decoder: is the architecture component that decompresses and reconstructs data from the encoded representation. The decoder is a mirror-like architecture of the encoder, but it incorporates upsampling layers instead of max pooling. Hence, the decoder blocks begin with an upsampling layer followed by two convolution layers. As we move from the block to the next, the number of filters is divided by two. The decoder, unlike the encoder unit, recalls specific convolution layers from the encoder. It joins them together between the blocks.

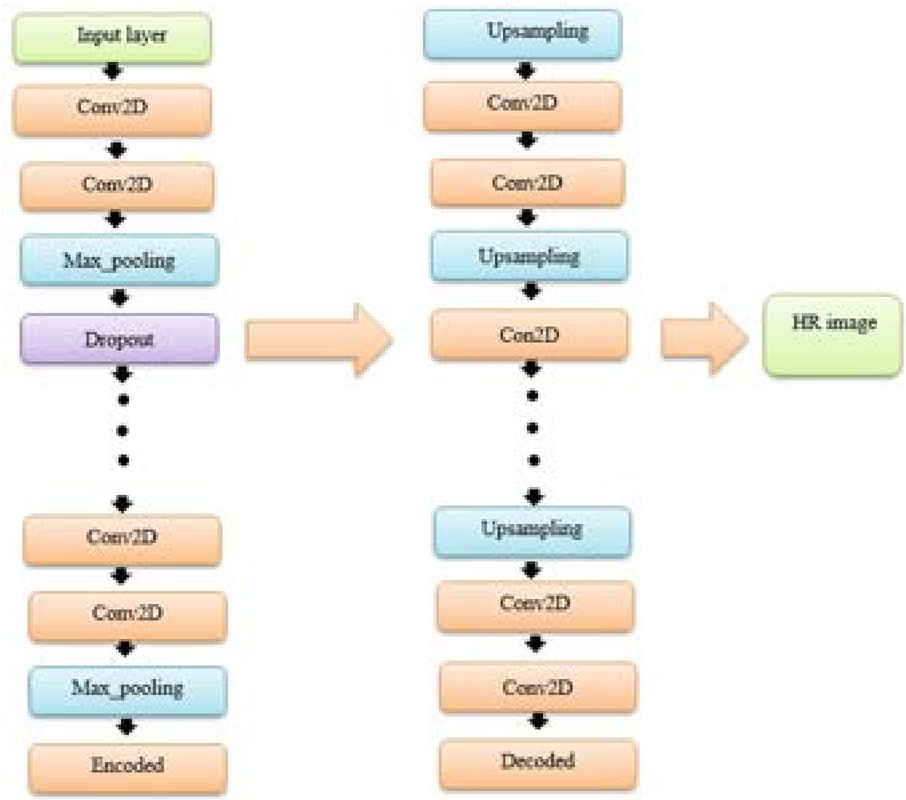

Fig. 2. The DCASR architecture.

3 Results and Discussion

3.1 Data Preparation

The data preparation step resulted in 1849 slices (251, 251) from the clipped scene (10793, 10793). Data augmentation generated mirror-like images from the initial 1849 ones. In total, we had 3698 images. The dataset was then divided into 2898 samples for training, 400 samples for testing, and 400 samples for validation.

Before using the input image in the model, we have proceeded by lowering the resolution without changing the size to 1/4th of its size. Then, we scaled back to the original size using bicubic interpolation (Fig. 3). We have now the original high- resolution image to compare the output with and the low-resolution images to provide to the model after being resized (256, 256) to be compatible with the model architecture (Fig. 2) (Figs. 4 and 5).

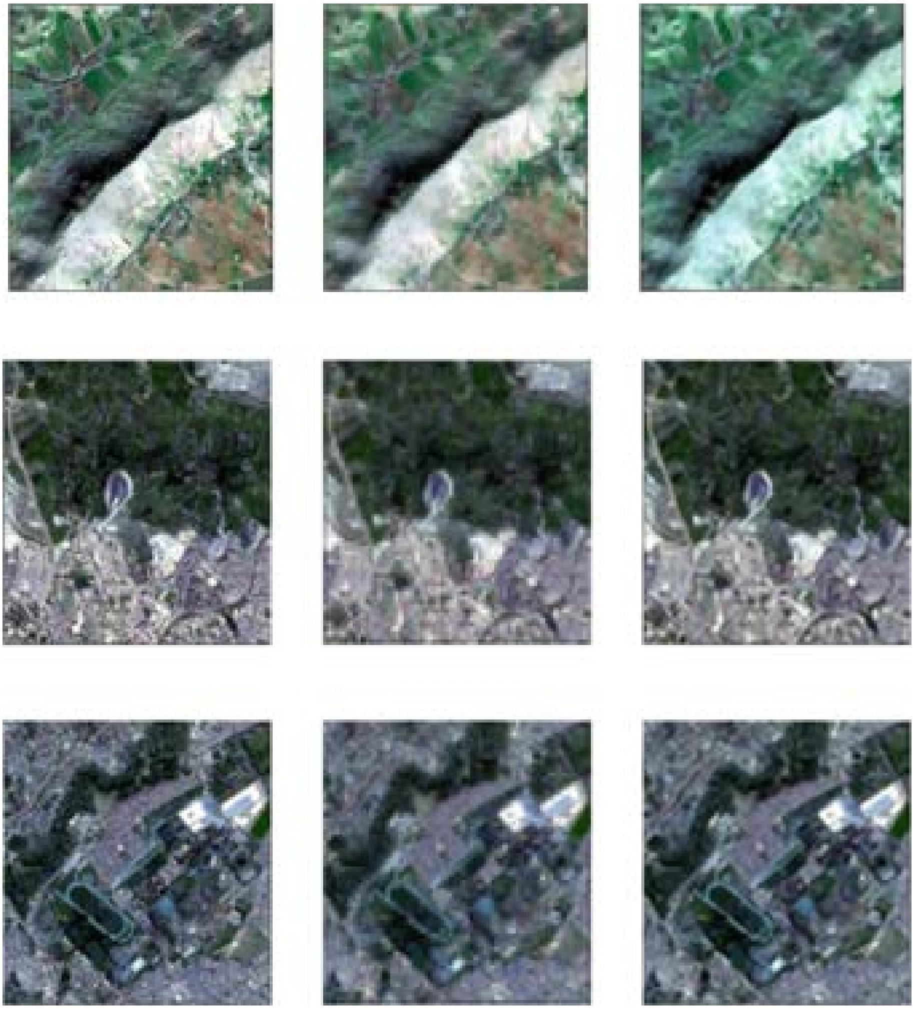

Fig. 3. The original images on the left; the degraded images in the middle, and our model selection on the right is.

3.2 DCASR Model

For validation purpose a quantitative evaluation of the results for our datasets are shown in (Table 3) and (Table 4). The accuracy is satisfying and the mean square error (MSE) is very low. We have also compared our model with other well-known architectures SRCNN model [20], VDSR [19], and the bi-cubic interpolation. All the models were run on the same dataset. The comparison was done based on tow metrics to judge the performance of the models, the peak signal-to-noise ratio (PSNR), and structural similarity (SSIM). As shown in (Table 3), our model recorded the highest PSNR and SSIM.

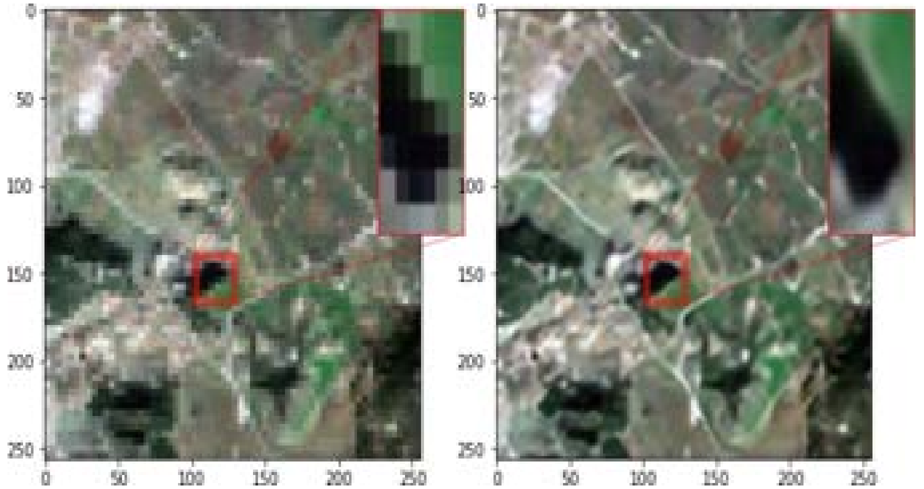

Fig. 4. The visual comparison between the resolution of degraded image (LR) and our HR result.

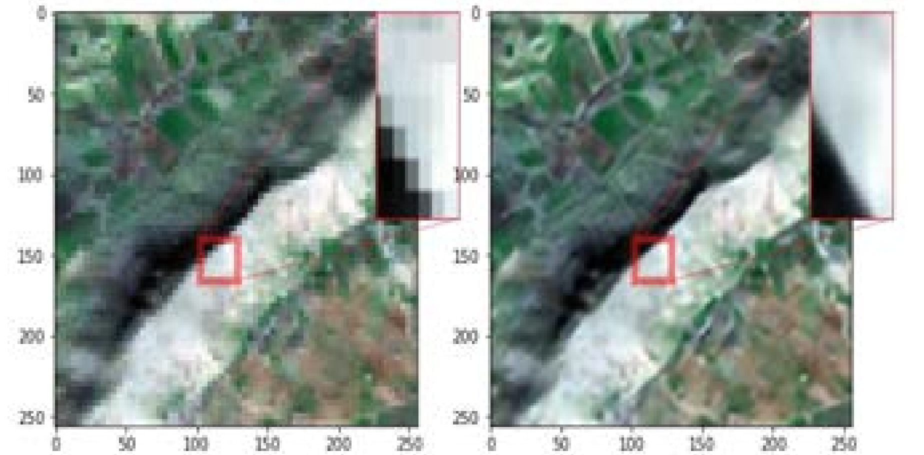

Fig. 5. The visual comparison between the resolution of degraded image (LR) and our HR result.

Table 3. Comparison of PSNR, and SSIM values for the three implemented models and bicubic interpolation. In bold the value indicating the best performing model

	SRCNN	Bi-cubic	VDSR	Ours
PSNR	30.98	29.56	31.68	31.94
SSIM	0.859	0.709	0.890	0.906

Table 4. Comparison The accuracy and MSE value for DCASR model.

Accuracy	MSE
0.9055	0.00637

DCASR model is training in our Dataset in 50 epochs the accuracy and the loss calculated (Fig. 6) (Fig. 7) were provided a high accuracy and a low loss that improve the performance of our proposed model

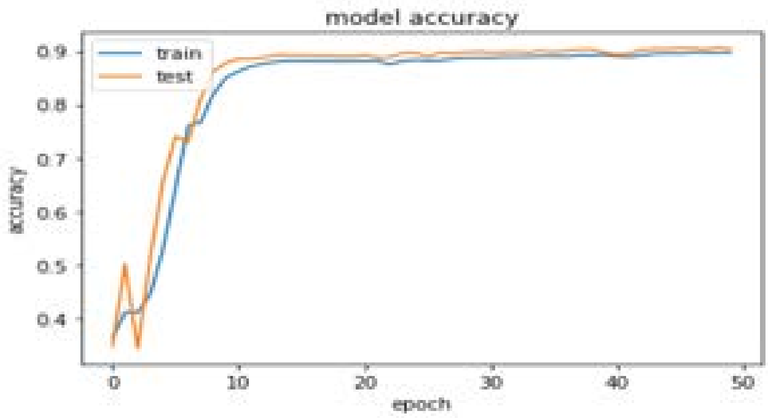

Fig. 6. Summarize history for accuracy.

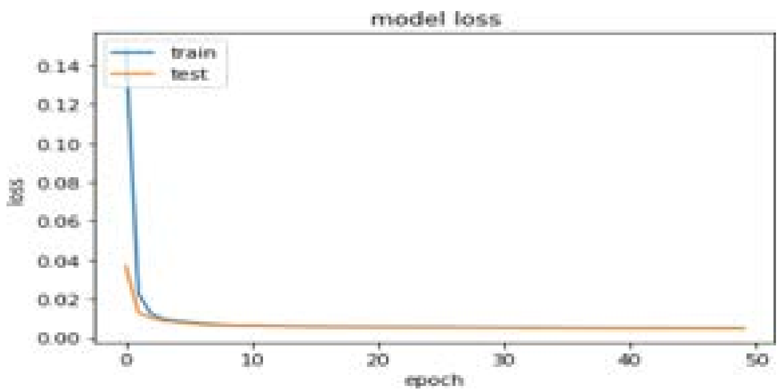

Fig. 7. Summarize history for loss.

We observe that 30 epochs are sufficient and that the accuracy and the loss do not change significantly after 30 epochs (Fig. 6) (Fig. 7). As a result, reducing the number of epochs reduces the model's complexity (the execution time).

4 Conclusion and Perspectives

The aim of this work was to explore the utility of applying super resolution methods on enhancing the spatial resolution of satellite images. This work presented a customized deep convolutional auto-encoder-based SR (DCASR) architecture. DCASR has proven its performance in applying it on Sentinel 2B MSI image slices. This was proven by the high accuracy and the low MSE (respectively 0.9055, 0.00637). Besides, our model outperformed other architectures (SRCNN, VDSR, and the bi-cubic interpolation). Hence our model could enhance the quality of the satellite images.

The use of Super_resolution in several domains implies challenges to improve the performance and reduce the learning time [27]; in this context, the perspectives proposed in this paper are as follows:

- Reduces the overall runtime of the training is important for improving performance. Batch normalization is a technique that achieves these goals in this context. In this regard, batch normalization techniques for super-resolution should be further investigated.
- In Super-resolution, network design architectures should be more thoroughly investigated because they affect how well any SR model performs.
- Another area of network design for quick and precise reconstruction of the HR image is the simultaneous use of low- and high-level representations to speed up the SR process.

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